

Notes: Gibbons and Waldman (QJE, 1999)
A Theory of Wage and Promotion Dynamics Inside Firms

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1. Big picture

Gibbons and Waldman develop a theory of wage and promotion dynamics inside firms. The paper is motivated by a set of empirical regularities from Baker, Gibbs, and Holmstrom (BGH) and related personnel studies that look difficult to reconcile with a simple competitive labor-market model.

The key idea is that careers inside firms are shaped by three forces at once:

- **Job assignment:** firms assign workers to jobs in a ladder according to the jobs in which the workers are most productive.
- **On-the-job human-capital acquisition:** workers become more productive as they gain labor-market experience.
- **Learning:** firms gradually learn about workers' underlying ability from noisy output signals.

The paper argues that no single force explains the facts well, but the combination explains many of them. Workers move up the job ladder as expected effective ability rises. Wages may fall when beliefs about ability are revised downward, but demotions can remain rare because job levels are coarse bins of productivity. Promotions come with large wage increases because promoted workers are selected from those whose expected ability has improved, yet the promotion raise is still small relative to the gap between average wages across job levels because workers at higher levels are typically older and have accumulated more human capital.

Interpretation: The paper turns puzzling personnel facts into equilibrium implications of a relatively standard model. The novelty is not a new exotic friction, but the integration of several familiar mechanisms.

2. Incremental contribution of the paper

2.1. Contribution relative to job-assignment models

Earlier job-assignment models emphasize comparative advantage and the scale-of-operations effect, but many are static. Gibbons and Waldman add a dynamic career structure. Assignment rules now generate predictions about promotions, demotions, wage changes, and fast-track careers.

2.2. Contribution relative to human-capital models

Human-capital models naturally generate rising wages with experience. On their own, however, they do not explain why real wage decreases occur, why demotions are rare, or why promotions are associated with special wage jumps. Gibbons and Waldman show that human-capital growth must be combined with learning and job assignment to fit the evidence.

2.3. Contribution relative to learning models

Learning models explain wage revisions as firms update beliefs about worker ability. But learning alone does not explain why jobs are organized as ladders or why promotions are tied to large but

incomplete wage changes. This paper makes beliefs about ability determine assignment to discrete job levels.

2.4. Contribution to the interpretation of BGH and Medoff-Abraham evidence

BGH argued that empirical facts inside a firm are hard to explain with standard models of human capital or learning considered separately. Gibbons and Waldman reinterpret the evidence: the facts do not reject standard ingredients; instead, they suggest these ingredients must be combined.

The paper also gives an interpretation of Medoff and Abraham's performance-evaluation results: performance evaluations may measure expected innate ability, not current productivity. Thus, evaluations can fall with experience within job level even though wages rise with experience.

2.5. Practical contribution

The paper offers a baseline framework for thinking about internal labor markets. It helps explain why wages, promotions, performance evaluations, and job titles are jointly determined rather than separable outcomes.

3. Evidence and measurement targeted by the model

This is a theory paper, not a new empirical paper. Its "data" are the empirical regularities established in prior personnel studies, especially BGH.

The main BGH facts the model tries to explain are:

1. **Real-wage decreases are not rare, but demotions are rare.**
2. **Wage increases and promotion rates are serially correlated.** Some workers appear to be on a fast track.
3. **Promotions are associated with large wage increases.**
4. **Promotion wage increases are small relative to average wage differences across job levels.**
5. **Large early wage increases within a level predict faster promotion to the next level.**

The paper treats these facts as a discipline for the model. A useful theory must explain not only why wages rise over careers, but also why job changes and wage changes have different dynamics.

Interpretation: Promotions are not merely pay raises, and wages are not merely attached to job titles. The model needs both wage dynamics and assignment dynamics.

4. Model structure and core equations

4.1. Workers

A worker has innate ability $\theta_i \in \{\theta_H, \theta_L\}$ and labor-market experience x_{it} . Effective ability is

$$\eta_{it} = \theta_i f(x_{it}), \quad (1)$$

where $f'(x) > 0$ and $f''(x) \leq 0$. Experience raises effective ability, but at a diminishing rate.

4.2. Jobs

There are three jobs, indexed by $j = 1, 2, 3$. If worker i is assigned to job j in period t , output is

$$y_{ijt} = d_j + c_j(\eta_{it} + \varepsilon_{ijt}). \quad (2)$$

The parameters satisfy $c_3 > c_2 > c_1 > 0$ and $0 < d_3 < d_2 < d_1$. Higher-level jobs have lower intercepts but higher returns to ability. This creates job-assignment thresholds.

Define η' as the effective ability level at which the worker is equally productive in jobs 1 and 2, and η'' as the level at which the worker is equally productive in jobs 2 and 3. The efficient assignment is:

- job 1 if $\eta_{it} < \eta'$;
- job 2 if $\eta' \leq \eta_{it} < \eta''$;
- job 3 if $\eta_{it} \geq \eta''$.

4.3. Wages and competition

Firms are identical and there is free entry. Workers and firms are risk neutral. Wages are set by spot-market competition, so a worker is paid expected output in the job that maximizes expected output. The paper imposes an infinitesimal moving cost to eliminate turnover in the symmetric-learning and full-information cases.

Interpretation: The model is competitive, but internal job levels matter because different jobs value ability differently. Promotions occur when expected effective ability crosses assignment thresholds.

5. Full-information benchmark

Under full information, each worker's innate ability is known at the start of the career. The worker is assigned to the job that maximizes current expected output, and the wage equals that expected output.

5.1. Assignment rule

The equilibrium assignment rule is direct:

$$\begin{aligned}\eta_{it} < \eta' &\Rightarrow j = 1, & w_{it} &= d_1 + c_1\eta_{it}, \\ \eta' \leq \eta_{it} < \eta'' &\Rightarrow j = 2, & w_{it} &= d_2 + c_2\eta_{it}, \\ \eta_{it} \geq \eta'' &\Rightarrow j = 3, & w_{it} &= d_3 + c_3\eta_{it}.\end{aligned}$$

Because $\eta_{it} = \theta_i f(x_{it})$ rises with experience, workers climb the job ladder over time. There are no demotions and no wage decreases in the full-information case.

5.2. What full information explains

The full-information version can explain several facts:

- **Serial correlation in wage increases:** high-ability workers receive larger wage increases in consecutive periods.
- **Fast tracks:** high-ability workers are promoted earlier and spend less time at each level before promotion.
- **Wage increases predict promotion:** a large wage increase signals a high-ability type whose effective ability will reach promotion thresholds sooner.
- **Promotion raises are small relative to level wage gaps:** higher job levels contain older, more experienced workers, so average cross-level wage gaps include accumulated human capital, not merely the promotion raise.

5.3. What full information cannot explain

The full-information model cannot explain real-wage decreases, because effective ability only rises with experience. It also does not fully match the timing and size of wage increases around promotion. Learning is needed.

Interpretation: Job assignment plus human capital creates a career ladder, but without learning there is no force that can move wages downward.

6. Symmetric learning and main predictions

6.1. Learning environment

In the symmetric-learning model, all firms and the worker begin with a common prior that the worker is high ability with probability p_0 . They observe noisy output and update beliefs. Let expected innate ability in period t be

$$\theta_{it}^e = E(\theta_i \mid z_{it-x}, \dots, z_{it-1}), \quad (3)$$

where z is normalized output. Expected effective ability is

$$\eta_{it}^e = \theta_{it}^e f(x_{it}). \quad (4)$$

Wages and job assignments are now based on η_{it}^e , not actual η_{it} .

6.2. Why wage decreases can occur without many demotions

Learning can revise expected innate ability downward. If the downward revision is large enough, expected effective ability can fall even though experience increases. Thus wages may decrease.

Demotions remain rare because assignment thresholds are discrete. A worker can suffer a wage cut within the same job level without crossing the lower threshold needed for demotion. This is the central mechanism explaining the coexistence of wage decreases and rare demotions.

6.3. Why promotions have large wage increases

Promotion wage increases are large because of two forces:

- **Assignment-slope effect:** higher jobs value ability more strongly, so crossing the threshold into a higher job raises the marginal value of effective ability.
- **Selection effect:** workers promoted in a given period are those whose expected innate ability has increased especially strongly.

This improves on the full-information version. In the full-information case, the promotion raise is driven mainly by assignment. In the learning case, the promotion raise also reflects good news about ability.

6.4. Why early wage increases predict fast promotion

A large early wage increase is evidence that beliefs about innate ability have improved. Since higher expected innate ability makes expected effective ability grow faster, the worker reaches the next promotion threshold sooner.

6.5. Why promoted workers come from high but not only top wages

The model predicts that promotion probability increases with the worker's percentile in the lower job's wage distribution. However, promotions need not come only from the very top, because learning can cause large jumps in expected effective ability.

Interpretation: Learning makes the wage ladder noisy enough to match the data, but the job ladder remains structured enough to keep demotions rare.

7. Performance evaluations, limitations, and alternative theories

7.1. Performance evaluations

Medoff and Abraham find that wages rise with experience within job levels, but performance evaluations do not rise and may even fall. Gibbons and Waldman explain this by interpreting performance evaluations as measures of expected innate ability, rather than current productivity.

Within a job level, more able workers are selected out through promotion. Thus older workers who remain in the same level may have lower expected innate ability on average, even though their wages are higher because they have accumulated more human capital.

7.2. Facts the model does not capture

The authors explicitly discuss three important facts that the symmetric-learning model does not fully explain:

- **Nominal wage rigidity:** real wage decreases occur, but nominal wage decreases are almost nonexistent in BGH.
- **Cohort effects:** entry conditions appear to affect wages many years later.
- **Green-card effect:** conditional expected wage increases are negatively related to initial wage within a job level in a way that may be stronger than the model predicts.

They suggest that nominal wage rigidity, hiring costs, aggregate shocks, and administrative wage rules may be needed to explain these patterns.

7.3. Alternative theories

The paper compares its model with three alternatives:

- **Asymmetric learning:** can explain large promotion wage jumps through signaling, but struggles with several BGH patterns that require heterogeneous wages among workers with the same experience and promotion history.
- **Insurance models:** can generate wage smoothing and rising profiles, but downward-rigid contracts conflict with frequent real wage decreases.
- **Tournament models:** can explain promotion incentives, but do not naturally explain the broad joint pattern of wage decreases, wage serial correlation, promotion timing, and promotion wage jumps.

Interpretation: The symmetric-learning model is not final, but it is a stronger baseline than models built on one mechanism alone.

8. Extracted source panels from the paper

The article contains no conventional empirical tables or standalone plotted figures. To keep the notes faithful to the paper and still directly extract visual material, the panels below reproduce key source excerpts from the original article: the abstract, the empirical facts, the model equations, propositions, a central corollary, limitations, and the conclusion.

A THEORY OF WAGE AND PROMOTION DYNAMICS INSIDE FIRMS* ROBERT GIBBONS AND MICHAEL WALDMAN We show that a framework that integrates job assignment, human-capital acquisition, and learning captures several empirical findings concerning wage and promotion dynamics inside firms, including the following. First, real-wage decreases are not rare but demotions are. Second, wage increases are serially correlated. Third, promotions are associated with large wage increases. Fourth, wage increases at promotion are small relative to the difference between average wages across levels of the job ladder. Fifth, workers who receive large wage increases early in their stay at one level of the job ladder are promoted quickly to the next. I. INTRODUCTION	define a single job ladder consisting of eight distinct levels. Throughout our discussion of their evidence we mean by a promotion or a demotion a movement up or down this job ladder. It would be interesting to know how many of these promotions and demotions involve changes only in job title, not in job content, but the data shed no light on this. The BGH findings we concentrate on are the following. First, real-wage decreases are not rare, while demotions are very rare. Second, there is significant serial correlation in both wage increases and promotion rates (the latter is sometimes referred to as a “fast track”). Third, promotions are associated with large wage increases. Fourth, wage increases at promotion are small relative to the difference between average wages across levels of the job ladder. Fifth, workers who receive larger wage increases early in their stay at one level of the job ladder are promoted more quickly
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(a) Title and abstract: the paper’s claim that job assignment, human-capital acquisition, and learning jointly explain wage and promotion facts.

(b) The BGH empirical facts: wage decreases, rare demotions, serial correlation, promotion raises, and fast-track patterns.

Read: Panel (a) states the research agenda. Panel (b) lists the facts the theory must match. These facts are not isolated; they jointly restrict what a credible model of internal labor markets can look like.

Economic message: The paper is designed as a unification exercise. The value of the model is measured by whether it can match a broad pattern of personnel facts simultaneously.

erations determine wage and promotion dynamics. As a benchmark, in this section we analyze this model under full information; the following section provides an analysis under symmetric learning. The technology of production we consider throughout the paper is similar to that investigated in Sattinger [1975], Rosen [1982], and Waldman [1984a]: jobs are ranked in terms of the extra value produced by a worker of greater ability. Sattinger, Rosen, and Waldman consider this technology in one-period models characterized by full information and no human-capital acquisition, while our focus is on multiple periods, human-capital acquisition, and (in later sections) learning.

A. The Model

There is free entry into production. All firms are identical, and the only input is labor. A worker's career lasts for T periods, $T \geq 5$ (an unusual assumption which we motivate below). In each period, labor supply is fixed at one unit for each worker. Worker i 's *innate ability* is denoted θ_i and can be either high or low: $\theta_i \in \{\theta_H, \theta_L\}$. A worker's *effective ability* is a function of the worker's innate ability and the worker's labor-market experience. Let t denote calendar time and x_{it} the worker's labor-market experience prior to period t (i.e., for a worker in his or her first period in the labor market, prior experience x_{it} equals zero). We assume that worker i 's effective ability in period t is given by

$$(1) \quad \eta_{it} = \theta_i f(x_{it}),$$

where $f' > 0$ and $f'' \leq 0$.

A firm consists of three different jobs, denoted 1, 2, and 3. If worker i is assigned to job j in period t , then the worker produces

$$(2) \quad y_{ijt} = d_j + c_j(\eta_{it} + \epsilon_{ijt}),$$

where d_j and c_j are constants known to all labor-market participants and ϵ_{ijt} is a noise term drawn from a normal distribution

tions; see footnote 7).

B. The Full-Information Benchmark

We now analyze the benchmark case of full information. That is, each worker's innate ability (θ_i) is common knowledge at the beginning of the worker's career. If innate ability is common knowledge, then in each period a worker is assigned to the job that maximizes the worker's expected output and paid a wage equal to that expected output.³ Proposition 1 characterizes the assignment rule in this case. We use w_{it} to denote the wage paid to worker i in period t in equilibrium. Proofs are given in the Appendix.

PROPOSITION 1. Suppose that each worker's innate ability is common knowledge at the beginning of the worker's career. Then job assignments and wages are given by (i) through (iii):

- (i) If $\eta_{it} < \eta'$, then worker i is assigned to job 1 in period t and earns the wage $w_{it} = d_1 + c_1\eta_{it}$.
- (ii) If $\eta' \leq \eta_{it} < \eta''$, then worker i is assigned to job 2 in period t and earns the wage $w_{it} = d_2 + c_2\eta_{it}$.
- (iii) If $\eta_{it} \geq \eta''$, then worker i is assigned to job 3 in period t and earns the wage $w_{it} = d_3 + c_3\eta_{it}$.

3. Throughout the analysis we assume that if a firm is indifferent between

(b) Proposition 1: full-information assignment and wage rule.

(a) Core model primitives: effective ability and output by job level.

Read: Panel (a) shows how effective ability is generated by innate ability and experience, and how job output depends on job-specific intercepts and slopes. Panel (b) gives the full-information assignment rule.

Economic message: The model creates a job ladder because higher jobs value ability more. Promotions are threshold-crossing events in expected effective ability.

Let θ_{it}^e denote the expected innate ability of worker i in period t : $\theta_{it}^e = E(\theta_i | z_{it-x}, \dots, z_{it-1})$. From θ_{it}^e we can compute the expected effective ability of worker i in period t :

$$(3) \quad \eta_{it}^e = \theta_{it}^e f(x_{it}).$$

Given η_{it}^e , job assignment and wage determination proceed as in the full-information case: in each period a worker is assigned to the job that maximizes the worker's expected output and is paid a wage equal to that expected output.

PROPOSITION 2. Suppose that learning is symmetric and that at the beginning of a worker's career the worker is known to be of innate ability θ_H with probability p_0 and of innate ability θ_L with probability $(1 - p_0)$. Then job assignments and wages are given by (i) through (iii):

- (i) If $\eta_{it}^e < \eta'$, then worker i is assigned to job 1 in period t and $w_{it} = d_1 + c_1 \eta_{it}^e$.
- (ii) If $\eta' \leq \eta_{it}^e < \eta''$, then worker i is assigned to job 2 in period t and $w_{it} = d_2 + c_2 \eta_{it}^e$.
- (iii) If $\eta_{it}^e \geq \eta''$, then worker i is assigned to job 3 in period t and $w_{it} = d_3 + c_3 \eta_{it}^e$.

Proposition 2 says that wages and job assignments are now determined by expected effective ability, whereas in the full-

(a) Proposition 2: under symmetric learning, assignment and wages depend on expected effective ability.

there exists a positive frequency of wage decreases but fewer (possibly no) demotions. Any demotions that occur are associated with wage decreases.

In the full-information case, a worker's effective ability increases each period, so in equilibrium there are no wage decreases. With symmetric learning, wage decreases occur because a worker's expected effective ability does not necessarily increase every period: because of the learning process, a worker's expected innate ability can fall substantially from one period to the next; if this decrease is sufficiently large, it will dominate the increase in expected effective ability due to human-capital accumulation.¹⁰

Our argument explains not only a positive frequency of wage decreases, but also why wage decreases are a minority of the observations, although not rare, while demotions are very rare. Roughly half the workers will experience a decrease in expected innate ability, so if human-capital acquisition is positive but small, then many of these will also experience a decrease in expected effective ability. The result is that wage decreases are a minority of the observations, but may not be rare. To see why demotions might be very rare or nonexistent, consider all workers assigned to job 2 in period t , and suppose that $\eta'' - \eta'$ is large. All workers for whom expected effective ability falls will experience a

(b) Corollary 3 and explanation: learning can generate wage decreases while demotions remain rare.

Read: Panel (a) replaces true effective ability with expected effective ability. Panel (b) shows the crucial prediction: after enough experience, wage cuts can occur because beliefs fall, but demotions are less frequent because the worker may remain within the same job-level band.

Economic message: The difference between wage adjustment and job adjustment is central. Wages are a continuous signal of expected productivity; job titles are discrete assignments.

discuss three findings from the BGH study that are not captured by our model. The first empirical finding inconsistent with our model involves nominal wage rigidity. Although BGH find a significant percentage of real wage decreases, nominal wage decreases are almost nonexistent in their sample—fewer than 200 observations in a sample of almost 70,000. (See Kahn [1997] and Card and Hyslop [1997] for other evidence that nominal wage decreases are rare.) Our model provides no rationale for why this should be the case, but an extension of our model does match some related evidence from BGH, as follows.

Suppose that one were to add to our model a constraint that firms could not offer nominal wage decreases and a hiring cost so that firms would not have an incentive to fire a worker whenever the worker's wage exceeded the worker's marginal product.¹⁵ In a model with these two new features, if the information revealed about the worker in period t is very negative, then in period $t + 1$ the firm would be constrained to pay a wage higher than would be predicted by our original model. Further, since the actual wage in $t + 1$ is too high, the wage increase in $t + 2$ should be very low. BGH find evidence in this spirit: the probability of a zero nominal increase this year, given a zero nominal increase last year, is two

(a) Limitations: nominal wage rigidity, cohort effects, and related BGH facts not captured by the baseline model.

VI. CONCLUSION

Recently, several empirical studies of careers in organizations have found results that are inconsistent with theoretical models based on any single factor such as human-capital acquisition or learning. In this paper we developed a model that integrates job assignment, on-the-job human-capital acquisition, and learning. We demonstrated that a model that combines these elements captures many of the findings in this recent literature. Our conclusion is not that our symmetric-learning model provides a perfectly accurate representation of careers in organizations. Rather, we find the results encouraging, and feel that our model offers a basic framework upon which to build more accurate models.

We have interpreted our model as applying to wage and promotion dynamics within firms. But our model assumes that wages are determined by spot-market competition and that an infinitesimal moving cost eliminates mobility between firms (although see footnote 2). Thus, if we added exogenous reasons for

¹⁷ Kwon's (1998) promising paper begins to capture the BGH findings using

(b) Conclusion: the model is a framework to build on, not a perfect description of internal careers.

Read: Panel (a) identifies patterns outside the reach of the baseline model. Panel (b) emphasizes that job assignment, human-capital acquisition, and learning are building blocks for future theories.

Economic message: The contribution is deliberately partial but constructive. The paper explains

a large set of facts, then clearly marks where additional mechanisms are needed.

9. Short summary

Gibbons and Waldman build a dynamic model of careers inside firms. Workers differ in innate ability, accumulate human capital, and are assigned to jobs that value ability differently. Under full information, workers climb a job ladder and wages rise, but the model cannot explain wage decreases. With symmetric learning, firms update beliefs about ability. A negative belief revision can reduce wages, while the worker may remain in the same job level, explaining why real-wage decreases are observed but demotions are rare.

The model also explains why wage increases are serially correlated, why promotion rates can display fast-track patterns, why promotion raises are large, why promotion raises are smaller than average wage gaps across job levels, and why workers with large early wage increases are promoted faster.

10. Conclusion

The paper's central lesson is that internal labor markets cannot be understood by studying wages, promotions, human capital, and learning separately. Wages respond continuously to expected productivity; promotions respond discretely to job-assignment thresholds; learning makes both dynamics stochastic; and human capital makes careers trend upward. The combination provides a coherent explanation for many facts that seem puzzling in isolation.

The model is not complete. It does not fully explain nominal wage rigidity, cohort effects, or all administrative features of firm pay systems. But it gives a disciplined baseline for future models that add asymmetric learning, incentives, and internal wage-setting rules.